

PREPARED FOR

T6N, R4W, Mount Diablo Meridian

COORDINATES 38.3459°N 122.3047°W WGS84	ELEVATION 66 ft NAVD88 · MSL	STATE PLANE CA 2 (0402) · N 1,887,922.81 E 6,474,289.05 (US ft, NAD83)	FEMA ZONE Zone X 06055C0510FF	DECLINATION 13.0° E WMM 2026
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APN 038050009000 | ZONING AP | ACRES 0.481 ac

BEST GNSS WINDOW Mon, Jul 13 · 6:00 AM – 10:00 AM | PDOP 0.99 · 34 sv | NEAREST CORS
P261 · 14.1 mi ✓

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Site file · Google Earth
Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First
St, Napa, CA 94559

NEAREST CORS
P261 · 14.1 mi

GENERATED
July 13, 2026

ASSESSOR
Napa County Assessor · 1127 First St,
Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com



This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.

SITE & SIGNAL REPORT

COORDINATES	ELEVATION	MAGNETIC DECLINATION	FEMA FLOOD ZONE
38.3459, -122.3047	66 ft	13.0° E	X
WGS84 / NAD83	Above MSL (approx)	Compass reads 13.0° too far W	AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

IONOSPHERIC CONDITIONS

PEAK KP INDEX

4.7

AVG KP

2.4

SOLAR FLUX F10.7

138 sfu

TEC (VTEC) ⚠ DEGRADED

18.4 TECU

-5.2 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.4). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Mon, Jul 13	0.99	6:00 AM – 10:00 AM	34	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.

🔥 WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 13, 2026) · Conditions may change – verify before fieldwork

✖ GPS CONSTELLATION HEALTH

⚠ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.8–0.8 across survey window
PDOP remains ≤ 0.8 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

Monday, Jul 13

Overcast

9 mph

GO

High 93°F

Low 59°F

Wind max 9 mph

Gusts 12 mph

Precip 0.00"

Sunrise 5:56 AM

Sunset 8:33 PM

Waning Crescent 2%

GNSS: Avg PDOP **0.99** · Best 4-hr window 6:00 AM – 10:00 AM (PDOP 0.94, 34 sats avg)

GNSS / DOP — HOURLY

	PDOP	SATS	RATING
6:00 AM	0.84	36 sv	IDEAL
7:00 AM	0.95	34 sv	IDEAL
8:00 AM	0.95	33 sv	IDEAL
9:00 AM	1.00	33 sv	GOOD
10:00 AM	1.01	29 sv	GOOD
11:00 AM	1.11	25 sv	GOOD
12:00 PM	1.03	26 sv	GOOD
1:00 PM	1.08	26 sv	GOOD
2:00 PM	0.93	32 sv	IDEAL
3:00 PM	0.97	33 sv	IDEAL
4:00 PM	0.98	33 sv	IDEAL
5:00 PM	0.99	33 sv	IDEAL
6:00 PM	1.02	30 sv	GOOD
7:00 PM	0.99	32 sv	IDEAL



GO — GOOD CONDITIONS

Partly cloudy | Wind: 9.4 mph (gusts 11.6 mph) | Precip: 0" | Temp: 59°–93.2°F



DRONE WIND ANALYSIS



GO Peak: 9.4/11.6 mph S

* Sunrise: 5:56 AM Sunset: 8:33 PM fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

SUSTAINED / GUSTS mph <=turbulence

	SUSTAINED / GUSTS mph	DIR
5 AM	2/4 mph	SE
6 AM	0/3 mph	W
7 AM	3/5 mph	E
8 AM	2/2 mph	SSW
9 AM	2/3 mph	SE
10 AM	4/5 mph	SSE
11 AM	5/5 mph	S
12 PM	5/6 mph	S
1 PM	5/5 mph	S
2 PM	9/10 mph	S
3 PM	7/8 mph	S
4 PM	8/9 mph	SSW
5 PM	9/11 mph	SSW
6 PM	9/11 mph	SSW
7 PM	9/12 mph	S
8 PM	8/11 mph	S

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ► FREE TO FLY

Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

FIELD PLANNING REPORT

GNSS SURVEY OPERATIONS | NAPA COUNTY, CA

Report Generated: Monday, July 13, 2026

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VERDICT: GO

Rationale: PDOP geometry is excellent (0.99 avg, 0.94 optimal window) with 34-satellite constellation available. Weather is stable (overcast, light winds, no precipitation). Ionospheric conditions are moderate with slightly depressed TEC; standard mitigation (longer static sessions) is sufficient. All constraints support survey execution.

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BEST SURVEY WINDOW

Monday, July 13, 2026 — 06:00 AM to 10:00 AM PDT

Reason: Peak geometry window with PDOP 0.94 and 34-satellite availability (GPS+GLONASS+Galileo+BeiDou multi-constellation). This 4-hour window minimizes ionospheric delay impact and ensures rapid, high-precision position fixes. Begin setup by 05:45 AM to capture first epoch at 06:00 AM.

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BEST 4-HOUR STATIC GNSS SESSION

06:00 AM – 10:00 AM PDT, Monday, July 13

Station occupancy: Minimum 120 minutes (2 hours) per setup given moderate ionospheric activity (Kp 4.7, TEC depressed 5.2 TECU below median). Elevation mask: 13° (terrain-optimized). Expected baseline precision: ±0.5–1.0 cm horizontal at 1σ. Receiver will auto-exclude PRN 11 (unhealthy); PDOP already reflects reduced constellation.

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DRONE OPERATIONS ASSESSMENT

FAVORABLE for visual observation/reconnaissance only. Weather permits flight operations (9 mph max wind, clear visibility, no precipitation). Confirm airspace clearance with FAA prior to deployment. No GNSS-dependent drone navigation required for this survey window. If drone carries RTK module, expect standard positioning accuracy within moderate ionospheric conditions; avoid drift corrections beyond 10 km from base station.

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FIELD SAFETY WARNINGS

Standard precautions apply:

- Heat advisory: Peak 93°F; maintain hydration and UV protection.
 - PRN 11 currently unhealthy (UNUSUFN). Receivers auto-exclude; no manual action required.
 - Wildfire proximity: No active fires within range (all-clear).
 - Air quality: AQI 45 (Good); respiratory risk minimal.
 - Overcast conditions reduce direct-sun heating but maintain standard PPE protocols.
- =====

SUMMARY

Survey is cleared for execution on Monday, July 13, 2026. Optimal 4-hour geometry window 06:00–10:00 AM with 0.94 PDOP and 34-satellite constellation. Moderate ionospheric activity (Kp 4.7, TEC -5.2 TECU) requires standard mitigation: 2+ hour minimum occupation per static session. Weather is stable (overcast, 9 mph max wind, no precipitation). One GPS satellite (PRN 11) is unhealthy and auto-excluded; PDOP reflects active constellation. Site conditions are favorable (low multipath, 13° terrain mask). Field crew should begin setup

by 05:45 AM and plan 120-minute minimum occupancies to ensure precision targets under current ionospheric conditions.

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COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk

1127 First St, Napa, CA 94559

ASSESSOR / TAX OFFICE

Napa County Assessor

1127 First St, Napa, CA 94559

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
 NGS ↗	P261	HUNTERHILLCN2004	14.1 mi
 NGS ↗	P198	PETALUMAIRCN2004	17.5 mi
 NGS ↗	OHLN	OHLONE PARK	23.5 mi
 NGS ↗	P196	MEACHUMLFLCN2006	24.0 mi
 NGS ↗	CASR	SANTA ROSA CA	24.9 mi
 NGS ↗	P267	DIXONAVIATCN2005	26.3 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet ↗	JT9010	"13"	19.100m / 62.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9018	"13 E"	17.970m / 59.0ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9009	"13"	17.900m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9017	"13 D"	17.890m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9034	"20 A"	20.730m / 68.0ft NAVD88	Order	0.3 mi
 Datasheet ↗	JT9012	"13"	19.060m / 62.5ft NAVD88	Order	0.4 mi
 Datasheet ↗	JT9043	"25 A"	18.710m / 61.4ft NAVD88	Order	0.4 mi
	JT9016	"13 D"	18.200m / 59.7ft NAVD88	Order	0.5 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet					

FEMA FLOOD ZONE

X – AREA OF MINIMAL FLOOD HAZARD

Panel: 06055C0510FF

Effective: Sep 29, 2010

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
JT9010	13	38.348333	-122.302778	62.7	
JT9018	13 E	38.348333	-122.302778	59.0	
JT9009	13	38.346667	-122.301111	58.7	
JT9017	13 D	38.345000	-122.301111	58.7	
JT9034	20 A	38.345000	-122.309444	68.0	
JT9012	13	38.351667	-122.304444	62.5	
JT9043	25 A	38.340000	-122.302778	61.4	
JT9016	13 D	38.341667	-122.297778	59.7	

[Download .txt File](#)

USGS STREAM GAUGES – within 25 miles

STATION	DIST	LEVEL	FLOW	FLOOD STATUS
NAPA R NR NAPA CA #11458000	1.5 mi	2.37 ft	4.39 cfs	● Normal
NATHANSON C A SONOMA CA #11458600	9.3 mi	—	—	● Normal
SONOMA C A AGUA CALIENTE CA #11458500	10.4 mi	8.84 ft	2.73 cfs	● Normal
NAPA R NR ST HELENA CA #11456000	14.1 mi	1.43 ft	3.23 cfs	● Normal
SONOMA CREEK A KENWOOD CA #11458433	14.9 mi	3.93 ft	0.19 cfs	● Normal
PUTAH C NR WINTERS CA #11454000	16.8 mi	7.85 ft	578 cfs	— Unknown
SUISUN SLOUGH NR BENICIA CA #11455740	19.6 mi	13.03 ft	2,230 cfs	— Unknown
POPE C A WALTER SPRINGS CA #11453590	21.1 mi	—	—	— Unknown

USGS LAKES & RESERVOIRS – within 150 miles

STATION	DIST	LEVEL	FULL POOL	LINK
CLEAR LK A LAKEPORT CA #11450000	58.1 mi	5.03 ft	N/R	USGS ↗

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	60.6 mi	696.94 ft (-9 ft)	706 ft	36K af	67%
Folsom Lake	66.5 mi	448.16 ft (-18 ft)	466 ft	787K af	81%
Lake Oroville	93.4 mi	861.14 ft (-39 ft)	900 ft	2,858K af	81%
New Melones Lake	100.1 mi	1,020.15 ft (-68 ft)	1,088 ft	1,664K af	69%
Black Butte Lake	101.2 mi	—	228 ft	74K af	54%

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
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ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	64 mi	696.9 ft	695.01 ft	—	100%
Camanche	89 mi	230.3 ft	221.86 ft	—	104%
Englebright Lake-Pool	95 mi	525.1 ft	522.35 ft	—	101%
Farmington Dam-Pool	99 mi	124.9 ft	117.09 ft	—	107%
Black Butte-Pool	101 mi	457 ft	458.36 ft	—	100%
Don Pedro	137 mi	—	N/R	—	—

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
005N004W32A002M	7 mi	200 ft	87.71 ft BGS	2014-11-05	N100CACSTL
005N004W28R001M	7 mi	51 ft	41.90 ft BGS	1977-10-03	N100CACSTL
004N004W04C001M	8 mi	80 ft	11.20 ft BGS	1977-10-03	N100CACSTL
004N004W05B001M	8 mi	—	54.10 ft BGS	1977-10-03	N100CACSTL
005N004W31R001M	8.1 mi	295 ft	—	—	—
004N004W05D002M	8.3 mi	60 ft	14.60 ft BGS	1977-10-03	N100CACSTL

004N004W06B001M	8.3 mi	201 ft	21.86 ft BGS	2022-06-07	—
004N004W02L001M	8.8 mi	—	9.90 ft BGS	1977-10-02	N100CACSTL

🕒 AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The Napa County site taps the Cascade Formation (N100CACSTL), a Pliocene-age sequence of volcanic and sedimentary deposits that form a moderately productive aquifer in the southern Coast Ranges. This formation was deposited in a marine-to-fluvial setting and subsequently tilted and faulted, creating confined to semi-confined conditions. Water availability at this site is moderate, with measured depths to water ranging from 9.9 to 87.7 ft below ground surface, indicating seasonal and longer-term fluctuations typical of a stressed or variably recharged system.

FORMATION

Name	Cascade Formation
Age / Rock	Pliocene age; mixed volcanic, sedimentary, and pyroclastic rock with interbedded silt and fine sand
Extent	est. 300–500 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	51–295 ft (5–30 floors)
Typical Depth	100–150 ft (10–15 floors)
Recharge Source	Precipitation on formation outcrops and lateral flow from higher Cascade elevations in the north
Transit Time	est. 20–50 years to 100 ft depth

WATER QUALITY

pH	est. 6.8–7.4 (est. USGS/DWR)
Hardness	est. 250–450 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 350–650 mg/L (est. USGS/DWR)
Notes	No sample data — typical for formation; expect moderate hardness and TDS from mixed volcanic-sedimentary mineralogy

TREND

Trend	Mixed trend: shallow wells show stable to rising levels (1977–2022); deep well (295 ft) shows significant drawdown from 41.90 ft (1977) to 87.71 ft (2014), indicating increased pumping stress
Source	USGS NWIS and site historical records
Seasonal Swing	est. 10–25 ft annual swing based on depth-to-water variance across dataset

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ LOW SUBSIDENCE RISK — No known critically overdrafted basin

Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🕒 AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU

The area near 38.3459, -122.3047 in Napa County overlies the North Coast Ranges aquifer system (N100CACSTL), characterized by fractured metamorphic and igneous bedrock with limited clay-rich aquitards typical of subsidence-prone regions. The well-drained soil classification and absence from critically overdrafted groundwater basins indicate that groundwater extraction has remained within sustainable yields, preventing the excessive aquifer depletion and clay layer compaction that drives land subsidence. For surveyors working in this area, published benchmarks and GPS-derived elevations should be considered stable and reliable for both new surveys and historical comparisons.

SUBSIDENCE ASSESSMENT

Susceptibility	Low
Primary Mechanism	None - consolidated terrain
Annual Rate	<0.5 mm/yr (est. USGS)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in SGMA basin

BENCHMARK IMPLICATIONS

Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence
GNSS Recommendation	Standard benchmark verification adequate

SURVEY ACTION

Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)

Site Class	D (default)
Risk Category	II

WIND / SNOW / ICE / FROST (ASCE 7-22)

Basic Wind Speed	92 mph
Ground Snow Load	5.8 psf
Winter Wind (W2)	0.35
Ice Thickness	None
Frost Depth	12 in minimum

Est. from IECC Zone 3C — AFI data unavailable

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

AI CLIMATE SUMMARY — RESIDENTIAL DESIGN

This marine climate stays mild year-round with moderate moisture, so your primary envelope concern is air sealing and moisture management rather than extreme temperature swings—prioritize continuous vapor barriers and high-quality weatherproofing. Orient glazing south and west to capture passive solar gains during cooler months, but shade south-facing glass with overhangs or vegetation to prevent overheating in summer, and minimize north and east-facing glass to reduce heat loss and marine fog penetration. The single most effective passive strategy here is natural cross-ventilation: design operable windows on opposite facades to let sea breezes cool the interior during warm afternoons, dramatically reducing cooling loads in a climate where you rarely need mechanical AC.

AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU

This coastal marine site in IECC Zone 3C presents moderate structural demands driven primarily by wind exposure and modest snow loading, with well-drained gravelly soils anticipated to support conventional spread footings. Seismic parameters are currently unavailable (null values) and may warrant clarification with the local building official or USGS; consult a licensed engineer to confirm applicability of any seismic design provisions. Frost depth of approximately 12 inches suggests standard foundation depth planning for this region.

FOUNDATION

Type Conventional spread footing anticipated

Basis Well-drained Coombs gravelly loam with 2–5% slopes and Site Class D suggest stable bearing capacity; seismic design parameters are null and should be confirmed with local authority before finalizing. Consult a geotechnical engineer to confirm bearing capacity and frost depth requirements.

STRUCTURAL IMPLICATIONS

Seismic Seismic design category (SDC) and spectral parameters are null; ordinary structural detailing may be adequate pending confirmation from local building official. Request clarification on applicable seismic design standards for this location.

Wind Exposure Exposure C (open terrain with scattered obstructions) anticipated per ASCE 7-22 Risk Category II; 92 mph 3-second gust wind speed suggested for planning—verify site shielding and exposure classification with engineer.

Wind Standard wind load design anticipated; no hurricane or extreme coastal wind exposure indicated. Typical braced frame or shear wall systems likely adequate—consult engineer for final design wind pressures.

Snow Moderate ground snow load of 5.8 psf anticipated; standard roof design with typical snow load coefficients likely adequate. Winter wind factor $W2 = 0.35$ may warrant review of wind-on-snow scenarios per ASCE 7-22.

INVESTIGATIONS & INSPECTIONS

Geotech Required Standard geotechnical investigation likely adequate; recommend confirmation of bearing capacity, settlement potential, and frost depth at foundation design stage. Site Class D assumed for planning—field investigation and lab testing may be warranted.

Liquefaction Screen Low risk anticipated; site is not in a known subsidence zone and soils are well-drained. Liquefaction screening recommended only if seismic parameters are confirmed; consult engineer if SDC is determined to be moderate or high.

Special Inspections Not anticipated based on current null seismic parameters; confirm applicability once SDC is established. Standard construction inspection practices likely adequate for this wind-dominated design.

Code Edition ASCE 7-22 / IBC 2021 (planning reference only)

Site Class Note Site Class D assumed for planning purposes; geotechnical investigation recommended to confirm soil profile, bearing capacity, and seasonal groundwater elevation. Local building official should clarify seismic design applicability and any site-specific requirements.

SOILS — USDA SOIL SURVEY

MAP UNIT: Coombs gravelly loam, 2 to 5 percent slopes **SERIES:** Coombs

DRAINAGE CLASS: Well drained **HYDRO GROUP:** C

SOIL ORDER: Alfisols **SUBORDER:** Xeralfs

AI SOIL INTELLIGENCE — CLAUDE HAIKU

Coombs gravelly loam is a well-drained Alfisol (Xeralf) formed in alluvial and colluvial materials on gently rolling terrain in the Coast Range and similar semi-arid regions. The soil profile features a gravelly loam surface over a clay loam argillic (clay-enriched) substratum, typical of Mediterranean climates with dry summers. Surveyors and developers should anticipate moderate gravel content affecting excavation, good bearing capacity for shallow foundations on slopes under 5 percent, and a clay-rich substratum that may retain moisture and exhibit moderate shrink-swell behavior during seasonal wet-dry cycles.

Alluvial and colluvial parent material on foothill and bajada slopes; well-drained; scattered oak and chaparral vegetation.

DRAINAGE

Class Well drained

Seasonal Saturation None within 60 inches

Flooding Risk Minimal on 2–5% slopes

Ponding Risk None

ENGINEERING

Bearing Capacity 3,000–4,000 psf (shallow foundations)

Rating Good

Shrink-Swell Moderate — clay loam argillic substratum

Frost Action Low

Cut Suitability Good (gravel-rich material; stone pick required)

Fill Suitability Fair (clay substratum — moisture conditioning and compaction control required)

SEPTIC

Suitability Conditional

Notes Perc test required — clay loam substratum typically 18–36 inches deep; percolation rate may range 2–6 in/hr; adequate depth and drainage expected if subsoil layer remains permeable

HAZARDS

- Excavation

Moderate — 15–35% gravel content; equipment wear anticipated
- Expansive Clay

Low to Moderate — monitor substratum clay during wet season
- Erosion

Low on gentle slopes; gully and rill risk in disturbed areas and road cuts
- Other

Gravel content may interfere with utility trenching; seasonal drying of clay substratum may cause minor settlement in poorly compacted fills

Data sources: USGS National Water Information System · [waterdata.usgs.gov](#) · California DWR / CDEC · [cdec.water.ca.gov](#) · USBR RISE · [data.usbr.gov](#) · USACE CWMS · [water.usace.army.mil](#) · USGS NWIS Groundwater · [waterservices.usgs.gov](#) · USDA Web Soil Survey · [sdmdataaccess.nrcs.usda.gov](#) · CA DWR SGMA Bulletin 118 · [gis.water.ca.gov](#) · ASCE 7-22 Hazard Tool · [gis.asce.org](#) · USGS Design Maps · [earthquake.usgs.gov](#) · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision
Water levels current as of report run: Jul 13, 2026, 2:47 AM UTC
* Live pool elevation not reported by this reservoir — full pool design elevation shown

 OIL & GAS WELLS California — within 10mi radius

8

total

0

active

7

plugged

By status:

7

Plugged

1

Idle

By type: 8 DH

Lease / Well	Status	Type	Operator	Depth	Dist
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
McKenzie-Hazard Syndicate	Plugged	DH	McKenzie-Hazard Syndicate	—	7.4 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.6 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.8 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	9.5 mi
Capell Oil Corp.	Idle	DH	Capell Oil Corp.	—	9.7 mi
Cordelia Oil & Gas Co.	Plugged	DH	Cordelia Oil & Gas Co.	—	12 mi

AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well is a plugged Mobil Oil Corporation well 7.2 miles away, which is beyond any actionable distance threshold. No active producers, injection wells, or idle wells exist within relevant review distances.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

200 **10** **108**
total in search area within 5 mi within 25 mi

Site Name	Status	Commodities	Distance
Roderick Sand and Gravel Pit	Occurrence	Sand/Gravel	1.6 mi
Basalt Rock Co	Producer	Clay	2 mi
Cicero Pumice Quarry	Occurrence	PUM	2.9 mi
Mount George King Pumice Deposit.	Occurrence	PUM	2.9 mi
Basalt Rock Co.	Occurrence	DIT	3.1 mi
Mt George Pumice Deposit	Producer	PUM	3.1 mi
Wilson Pumice Quarry	Occurrence	PUM	3.7 mi
Unknown	Occurrence	Mercury	4.4 mi
Silva Pumice Quarry	Occurrence	PUM	4.7 mi
Silva Pumice Quarry	Producer	PUM	5 mi
Walker Pumice Quarry	Occurrence	PUM	5.1 mi
Walker Pumice Quarry	Producer	PUM	5.1 mi
Mountain	Occurrence	Mercury	5.4 mi
Unknown	Occurrence	Mercury	7.6 mi
Unknown	Occurrence	Stone/Crushed	7.8 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Prospect	Occurrence	Chromium	8.8 mi
Unnamed Location	Occurrence	Chromium	9 mi
Unknown	Occurrence	Chromium	9.1 mi
Bella Oak	Past Producer	Mercury	9.4 mi
Unknown	Occurrence	MG	9.7 mi
Serres Pit	Past Producer	Sand/Gravel	10.4 mi
Unnamed Location	Unknown	Sand/Gravel	10.5 mi
Conn Valley	Occurrence	Manganese	10.6 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

Mineral Resource Review – Site Planning Assessment Three active mineral producers operate within 5 miles of the project site. The nearest are Basalt Rock Co. (clay extraction, 2 miles) and Mt George Pumice Deposit (pumice production, 3.1 miles), followed by Silva Pumice Quarry (pumice, 5 miles). Plan for heavy truck traffic on local access roads during operating hours, dust generation from aggregate and clay operations, and potential noise from crushing or blasting activities at the pumice facilities. No inactive deposits or occurrences warrant Phase 1 environmental assessment based on proximity.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6 **6** **12**
Endangered Threatened Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
California tiger Salamander	<i>Ambystoma californiense</i>	Endangered
Conservancy fairy shrimp	<i>Branchinecta conservatio</i>	Endangered
Contra Costa goldfields	<i>Lasthenia conjugens</i>	Endangered
Soft bird's-beak	<i>Cordylanthus mollis</i> ssp. <i>mollis</i>	Endangered
Suisun thistle	<i>Cirsium hydrophilum</i> var. <i>hydrophilum</i>	Endangered
Vernal pool tadpole shrimp	<i>Lepidurus packardii</i>	Endangered
Alameda whipsnake (=striped racer)	<i>Masticophis lateralis euryxanthus</i>	Threatened
California red-legged frog	<i>Rana draytonii</i>	Threatened
Delta smelt	<i>Hypomesus transpacificus</i>	Threatened
Northern spotted owl	<i>Strix occidentalis caurina</i>	Threatened
Vernal pool fairy shrimp	<i>Branchinecta lynchi</i>	Threatened
western snowy plover	<i>Charadrius nivosus nivosus</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Planning Assessment - 38.3459, -122.3047 **ESA Section 7 Triggers & Planning Relevance:** Section 7 consultation will likely be triggered if the proposed project involves federal permitting, funding, or licensing (e.g., Army Corps Clean Water Act permits, USFWS authorizations). Given the site's location in the San Francisco Bay/Delta region, the most planning-relevant species are **California red-legged frog** and **vernal pool species** (tadpole shrimp, fairy shrimp), which typically indicate seasonal wetland/pond habitats requiring avoidance or mitigation, and **Delta smelt** if the project affects water diversions or quality. The **California tiger salamander** and **Contra Costa goldfields** warrant habitat assessment if grassland or seasonal pool features exist on-site. **Recommended Next Step:** Retain a qualified environmental consultant to conduct a Phase 1 biological survey and ESA screening to identify whether critical habitat occurs on or immediately adjacent to the project footprint, delineate seasonal wetlands/vernal pools, and determine whether project activities (grading, water management, vegetation removal) would trigger incidental take or habitat modification requiring Section 7 formal consultation or Section 10 incidental take permits. **For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.**

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

57 **5051** **2026-07-04**
Unique Species Total Observations Most Recent

COMMON NAME	SCIENTIFIC NAME	LAST OBS
American black nightshade	<i>Solanum americanum</i>	2026-05-29
Bearded Clover	<i>Trifolium barbigerum</i>	2026-05-15
bird's foot cliffbrake	<i>Pellaea mucronata</i>	2026-04-28
Blow Wives	<i>Achyrachaena mollis</i>	2026-05-20
blue oak	<i>Quercus douglasii</i>	2026-05-09
Bolander's Phacelia	<i>Phacelia bolanderi</i>	2026-04-27
box elder	<i>Acer negundo</i>	2026-05-18
Braun's Giant Horsetail	<i>Equisetum telmateia braunii</i>	2026-06-22
California bay	<i>Umbellularia californica</i>	2026-06-24
California beeplant	<i>Scrophularia californica</i>	2026-05-20
California black oak	<i>Quercus kelloggii</i>	2026-05-15
California buckeye	<i>Aesculus californica</i>	2026-06-02
California Dutchman's Pipe	<i>Aristolochia californica</i>	2026-05-26
California milkwort	<i>Rhinotropis californica</i>	2026-05-15
California mugwort	<i>Artemisia douglasiana</i>	2026-06-02
California poppy	<i>Eschscholzia californica</i>	2026-06-24
California Wild Rose	<i>Rosa californica</i>	2026-05-09
Canyon liveforever	<i>Dudleya cymosa</i>	2026-05-30
-	<i>Castilleja densiflora densiflora</i>	2026-05-09
clay mariposa lily	<i>Calochortus argillosus</i>	2026-05-15
coast live oak	<i>Quercus agrifolia</i>	2026-05-13
coastal woodfern	<i>Dryopteris arguta</i>	2026-05-20
common selfheal	<i>Prunella vulgaris</i>	2026-06-22
Diogenes' lantern	<i>Calochortus amabilis</i>	2026-05-15
distant phacelia	<i>Phacelia distans</i>	2026-05-15
-	<i>Downingia concolor concolor</i>	2026-05-15
Dwarf Brodiaea	<i>Brodiaea terrestris</i>	2026-05-15
Elegant Clarkia	<i>Clarkia unguiculata</i>	2026-05-16

COMMON NAME	SCIENTIFIC NAME	LAST OBS
goldback fern	<i>Pentagramma triangularis</i>	2026-05-20
Ithuriel's Spear	<i>Triteleia laxa</i>	2026-06-15
maroonspot calicoflower	<i>Downingia concolor</i>	2026-05-15
narrowleaf milkweed	<i>Asclepias fascicularis</i>	2026-06-25
narrowleaf mule-ears	<i>Wyethia angustifolia</i>	2026-05-15
Narrowleaf Willow	<i>Salix exigua</i>	2026-06-24
needleleaf navarretia	<i>Navarretia intertexta</i>	2026-05-15
northern California black walnut	<i>Juglans hindsii</i>	2026-06-24
orange bush monkeyflower	<i>Diplacus aurantiacus</i>	2026-05-29
Oregon Ash	<i>Fraxinus latifolia</i>	2026-06-02
Pacific Bleeding Heart	<i>Dicentra formosa</i>	2026-05-04
Pacific madrone	<i>Arbutus menziesii</i>	2026-07-04
Pacific poison oak	<i>Toxicodendron diversilobum</i>	2026-06-02
pineapple-weed	<i>Matricaria discoidea</i>	2026-05-20
Redwood Sorrel	<i>Oxalis smalliana</i>	2026-05-28
small-headed clover	<i>Trifolium microcephalum</i>	2026-05-15
Superb Mariposa Lily	<i>Calochortus superbus</i>	2026-06-11
thimble clover	<i>Trifolium microdon</i>	2026-05-15
tomcat clover	<i>Trifolium willdenovii</i>	2026-05-15
Toyon	<i>Heteromeles arbutifolia</i>	2026-06-19
trailing blackberry	<i>Rubus ursinus</i>	2026-06-24
Tree Clover	<i>Trifolium ciliolatum</i>	2026-05-15
valley oak	<i>Quercus lobata</i>	2026-06-24
wavy-leafed soap plant	<i>Chlorogalum pomeridianum</i>	2026-05-20
western sycamore	<i>Platanus racemosa</i>	2026-05-17
white brodiaea	<i>Triteleia hyacinthina</i>	2026-05-15
white-tipped clover	<i>Trifolium variegatum</i>	2026-05-15
Winecup Clarkia	<i>Clarkia purpurea</i>	2026-05-15
Yellow Mariposa Lily	<i>Calochortus luteus</i>	2026-06-11

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment Summary This site in the inner Coast Ranges supports a **mixed oak woodland and chaparral community**, dominated by blue oak and California black oak with understory elements of California bay, California buckeye, and native shrubs like California beeplant and phacelia. The presence of moisture-loving species (Braun's Giant Horsetail, California wild rose) alongside drought-adapted plants (dudleya, mariposa lily, cliffbrake) indicates **transitional terrain with seasonal water movement**, likely in draws or north-facing slopes with intermittent moisture. This mixed moisture regime suggests **variable soil drainage across the site—expect poorly drained pockets in valley areas and rapid drainage on slopes**—which will require careful grading design to avoid ponding during wet seasons while maintaining adequate water retention for oak regeneration. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

635	154	91	1439
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants
VENOMOUS SNAKES			
SPECIES		OBSERVATIONS	
Crotalus oreganus oreganus		635 obs	
VENOMOUS SPIDERS			
SPECIES		OBSERVATIONS	
Latrodectus hesperus		154 obs	
STINGING INSECTS			
SPECIES		OBSERVATIONS	
Dasymutilla aureola		91 obs	
Dasymutilla californica		91 obs	
Dasymutilla sackenii		91 obs	
TOXIC PLANTS			

SPECIES	OBSERVATIONS
<i>Toxicodendron diversilobum</i>	1100 obs
<i>Datura wrightii</i>	57 obs
<i>Datura stramonium</i>	57 obs
<i>Datura ferox</i>	57 obs
<i>Conium maculatum</i>	199 obs
<i>Urtica gracilis</i>	83 obs
<i>Urtica gracilis holosericea</i>	83 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Brief – Napa County Area Your highest-priority hazard is **poison oak** (*Toxicodendron diversilobum*)—extremely common in this region with nearly 1,500 observations, causing severe skin reactions on contact. Second concern is **Northern Pacific rattlesnakes** (*Crotalus oreganus oreganus*), which are present but predictable: wear closed-toe boots, watch where you step and place your hands, and avoid tall grass at dawn/dusk. Wear long pants and sleeves, use barrier creams on exposed skin, and thoroughly wash clothes and tools after fieldwork to prevent poison oak transfer; learn to identify its three-leaflet pattern year-round. Late spring through fall brings peak snake activity and poison oak oil potency, so schedule work in early morning when snakes are less active, and remember that poison oak is toxic even in winter dormancy. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

3	545ft	3		
Beam Paths	Closest Beam	Fixed P2P		
TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WBO61	WBO59	Fixed Point-to-Point	—	545ft
KYH37	WQAT627	Fixed Point-to-Point	—	927ft
WRCR329	WQAT627	Fixed Point-to-Point	—	1289ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

26	13	2	11			
Total Airways	Jet Routes	Victor Airways	Other Routes			
AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	317.1°	135.4°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.0°	127.3°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	161.4°	341.7°
T257	T-Route	Low	—	—	321.1°	140.5°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

8

Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Other	1	No	No	—
CFNR	Yard	1	No	No	—
CFNR	Mainline	1	No	No	VALLEJO
SMRT	Mainline	1	No	No	BRAZOS JCT.
NVRR	Mainline	1	No	No	MARTINEZ
USG	Other	1	No	No	—
CFNR	S	1	No	No	—
NVRR	Yard	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200

Total Observations

89

Total Species

64

Birds

5

Mammals

14

Reptiles

6

Amphibians

BIRDS (64 species)

COMMON NAME	Scientific Name	OBS
Wild Turkey	<i>Meleagris gallopavo</i>	10
Snowy Egret	<i>Egretta thula</i>	7
California Quail	<i>Callipepla californica</i>	5
Western Bluebird	<i>Sialia mexicana</i>	5
California Towhee	<i>Melospiza crissalis</i>	5
Great Egret	<i>Ardea alba</i>	4
Red-tailed Hawk	<i>Buteo jamaicensis</i>	4
Black Phoebe	<i>Sayornis nigricans</i>	4
Northern Flicker	<i>Colaptes auratus</i>	3
Cooper's Hawk	<i>Astur cooperii</i>	3
Ash-throated Flycatcher	<i>Myiarchus cinerascens</i>	3
Western Screech-Owl	<i>Megascops kennicottii</i>	3
Turkey Vulture	<i>Cathartes aura</i>	3
Mallard	<i>Anas platyrhynchos</i>	3
Bushtit	<i>Psaltiriparus minimus</i>	3

+ 49 additional species

MAMMALS (5 species)

COMMON NAME	Scientific Name	OBS
Coyote	<i>Canis latrans</i>	3
Black-tailed Jackrabbit	<i>Lepus californicus</i>	3
Mule Deer	<i>Odocoileus hemionus</i>	2
Eastern Fox Squirrel	<i>Sciurus niger</i>	1
Striped Skunk	<i>Mephitis mephitis</i>	1

REPTILES (14 species)

COMMON NAME	Scientific Name	OBS
Western Fence Lizard	<i>Sceloporus occidentalis</i>	10
Pacific Gopher Snake	<i>Pituophis catenifer catenifer</i>	7
Northern Pacific Rattlesnake	<i>Crotalus oreganus oreganus</i>	6
California King Snake	<i>Lampropeltis californiae</i>	3
Pond Slider	<i>Trachemys scripta</i>	2
Western Skink	<i>Plestiodon skiltonianus</i>	2

COMMON NAME	Scientific Name	OBS
Ring-necked Snake	<i>Diadophis punctatus</i>	1
Common Garter Snake	<i>Thamnophis sirtalis</i>	1
Western Yellow-bellied Racer	<i>Coluber constrictor mormon</i>	1
Pacific Ringneck Snake	<i>Diadophis punctatus amabilis</i>	1
Northwestern Pond Turtle	<i>Actinemys marmorata</i>	1
Common Sharp-tailed Snake	<i>Contia tenuis</i>	1
North American Racer	<i>Coluber constrictor</i>	1
Southern Alligator Lizard	<i>Elgaria multicarinata</i>	1

AMPHIBIANS (6 species)

COMMON NAME	Scientific Name	OBS
Pacific chorus frog	<i>Pseudacris regilla</i>	5
American Bullfrog	<i>Lithobates catesbeianus</i>	3
Foothill Yellow-legged Frog	<i>Rana boylei</i>	3
Western Toad	<i>Anaxyrus boreas</i>	2
Rough-skinned Newt	<i>Taricha granulosa</i>	1
California Giant Salamander	<i>Dicamptodon ensatus</i>	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment – 38.3459, -122.3047 This site supports a diverse mixed community characteristic of California's foothills and oak woodland transitional zones, with strong representation across birds, mammals, reptiles, and amphibians indicating mosaic habitat conditions. Species of regulatory concern present include the Foothill Yellow-legged Frog and California Giant Salamander, both sensitive to hydrologic alterations and water quality degradation, as well as Northwestern Pond Turtle, suggesting perennial or seasonal aquatic features warrant protection. The amphibian assemblage, particularly the diversity of salamanders and frogs alongside wetland-dependent reptiles (Pond Slider, garter snakes), indicates proximity to riparian corridors, springs, or persistent freshwater habitats that are critical for species viability. The overall species composition and research-grade documentation density suggest a site with relatively intact habitat structure and moderate disturbance levels, though the presence of introduced species (Eastern Fox Squirrel, American Bullfrog) indicates some legacy human influence; any proposed development should prioritize retention of aquatic features and riparian buffers to maintain this ecological productivity. For planning purposes only. Consult a qualified biologist for site-specific biological assessments.

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

* PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 · AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 · IDEAL SOLAR ARRAY PITCH

Optimal tilt = 38.3° (latitude) → 9/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97%	good
5/12	22.6°	97.6%	good
6/12	26.6°	98.2%	↪ excellent
7/12	30.3°	98.8%	↪ excellent
8/12	33.7°	99.3%	↪ excellent
9/12	36.9°	99.8%	↪ annual optimal
10/12	39.8°	99.8%	↪ excellent
12/12	45°	99%	↪ excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~\$108 value/yr). Over 25yr panel life: ~\$2710 total gain.

3 · SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	51	31	Minimize south glass
South glass (SHGC 0.6)	76	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	27	104	⚠ Minimize — afternoon load
Roof — dark (absorb 0.95)	236	301	Highest load surface
Roof — light (absorb 0.35)	87	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	62	79	Best roof option for heat
South wall — dark	94	69	
South wall — light	31	23	Light color = major savings
Hardscape — asphalt	181	301	⚠ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr-ft² additional load per 100sf of adjacent hardscape.
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 • ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 • THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 • TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.1	1.9	2.4	OK
30ft	All year	3.1	2.9	3.7	Zone 2 check
40ft	All year	4	3.8	5.2	△ verify
50ft	All year	4.8	4.6	7.3	△ verify
60ft	All year	5.6	5.5	9.3	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.

Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.

△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 • ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45

Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45

Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor

†**Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)

WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.

△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🔍 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Solar & Energy Planning for Your Home Your location gets about 6.6 hours of useful sun per day on average—think of it like having a reliable part-time solar worker showing up almost every day, though winter cuts that down to about 4 hours. Your roof's angle and color are surprisingly powerful: a dark roof soaks up heat like a black car left in the sun (301 BTU per hour per square foot in summer), while a light-colored roof reflects much of that away, cutting the heat gain down to about a third—and since your roof naturally tilts at roughly 38 degrees, that's actually the sweet spot for catching winter sun when you need it most. The tricky part is your west-facing windows: they're like an afternoon oven door, letting in twice as much summer heat as your south windows, so you'll want shading there—either with overhangs, shutters, or curtains you can close. Here's where nature helps: a big deciduous tree on the south side is your free air conditioner, blocking summer sun but dropping its leaves in winter to let warming light through (you can think of it as a living seasonal shade). One last concept to understand is thermal mass—basically, materials like concrete or ICF walls that absorb heat slowly and release it slowly, so your house doesn't swing wildly between hot and cold. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.



Technical Brief — for architects, engineers, and surveyors

Site Intelligence Briefing 38.34586°N, 122.30466°W | Napa County, California

Structural Design Drivers & Load Hierarchy

Seismic loading is absent from your dataset, wind and snow data are not available, and frost depth is unreported for this location. This data gap is the first problem: you cannot rank structural design drivers without wind speed (3-second gust or equivalent) and snow load parameters. However, the site falls within IECC Zone 3C (Marine) with an accumulated freezing index of 0°F-days, indicating a frost-free climate. This eliminates frost heave as a footing design constraint but does not tell you the governing lateral load (wind vs. seismic).

Before selecting structural system type or foundation depth, obtain site-specific wind speed data from ASCE 7 Section 26 for Napa County and confirm whether IBC seismic provisions apply to this coordinate. The absence of Quaternary faults within the search radius suggests low seismic hazard proximity, but that does not eliminate seismic design requirements under California Building Code—only that ground rupture risk appears minimal.

Geotechnical Foundation Considerations

The site is underlain by Coombs gravelly loam on slopes of 2 to 5 percent. This soil is classified as Hydrologic Group C (moderately high runoff potential) and is reported as well-drained. The texture designation is incomplete in your dataset; confirm whether "gravelly loam" means loam with gravel or gravelly clay loam, as this affects bearing capacity and shrink-swell potential. Well-drained classification is favorable for conventional shallow foundations and reduces risk of seasonal water table rise. No frost depth is reported; in IECC Zone 3C marine climates, frost depth is typically shallow (12–18 inches in northern California), but site-specific boring data should confirm this before setting footing elevation. Subsurface investigation via Phase I geotechnical boring is essential to confirm bearing capacity, measure actual groundwater depth, and verify soil gradation—especially given proximity to active mining operations that may have altered subsurface conditions.

Hazard Assessment & Constraints

No Quaternary faults plot within the search radius, and no subsidence basin data is available for this location, which are positive indicators. However, the site is located within active mining territory: three operating quarries exist within a 5-mile radius (Basalt Rock Co at 2 miles for clay; Mt George Pumice and Silva Pumice quarries at 3.1 and 5 miles for pumice). Pumice and basalt extraction generates blast vibration and ground noise; confirm whether air blast or vibration limits apply under Napa County grading or conditional use permits for those operations. No active or producing oil/gas wells exist within 10 miles (the nearest—Mobil Oil Corporation—is 7.2 miles distant and is plugged), eliminating subsurface pressure or induced seismicity risk. Wildfire hazard data is not provided; request State Responsibility Area (SRA) or local fire hazard severity zone designation for this coordinate.

Site Climate & Development Context

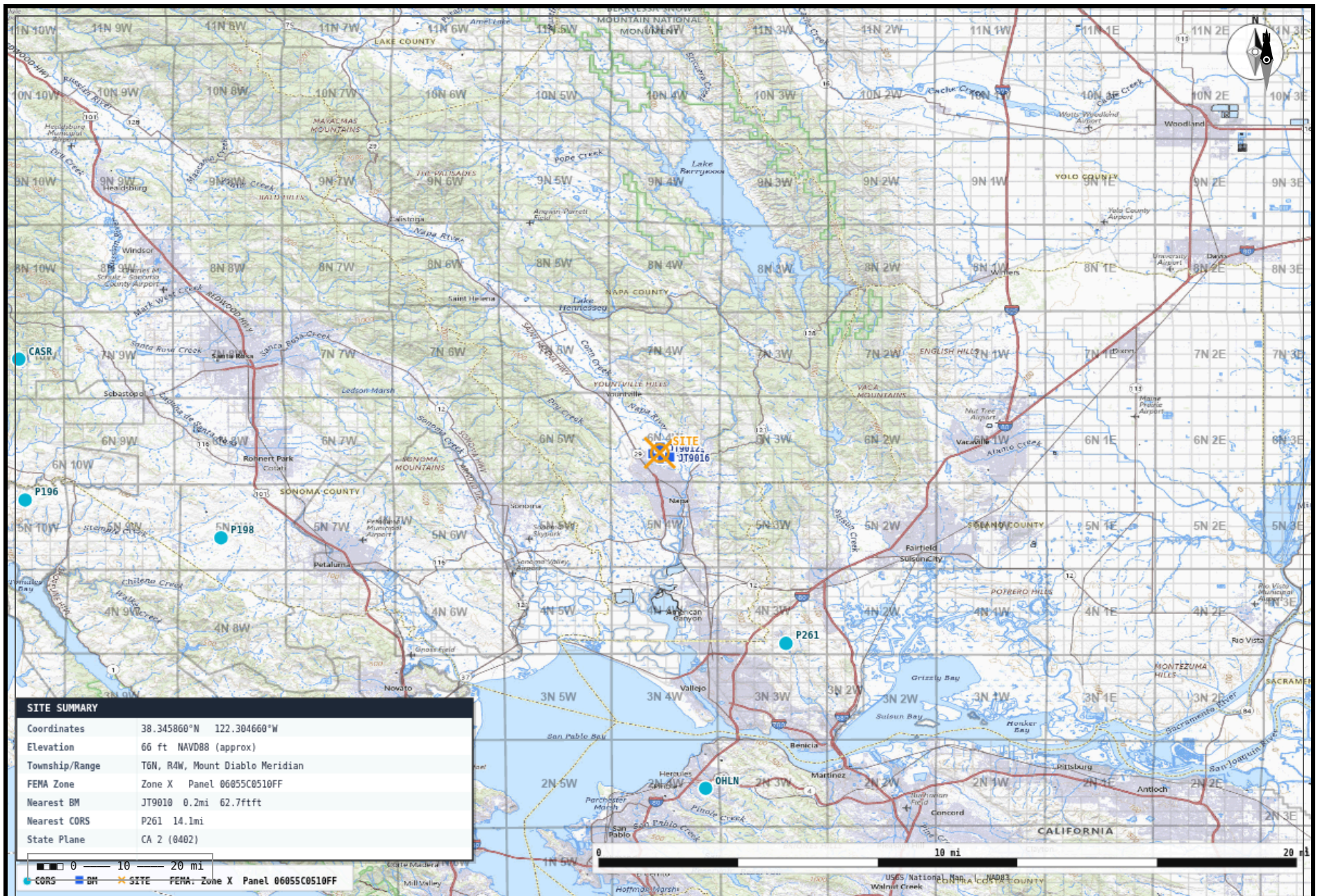
IECC Zone 3C (Marine) places this location in a mild, maritime climate with minimal heating degree days and no freezing risk—design envelope allows thin or uninsulated shallow foundations and reduces mechanical system complexity. The well-drained soil and low frost depth support conventional spread footing or slab-on-grade systems. Slope gradient of 2 to 5 percent is gentle; erosion and stormwater management should address Hydrologic Group C runoff characteristics but do not immediately flag slope stability concerns. Proximity to three active mineral extraction sites within 5 miles may introduce regulatory compliance requirements (conditional use review, vibration monitoring, setback distances) depending on site zoning and Napa County land use designations.

Immediate Action: Request site-specific wind speed data, seismic design category confirmation, and geotechnical boring results before finalizing structural system or foundation type; confirm mining operation setback distances and blast monitoring requirements with Napa County Planning.



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Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.

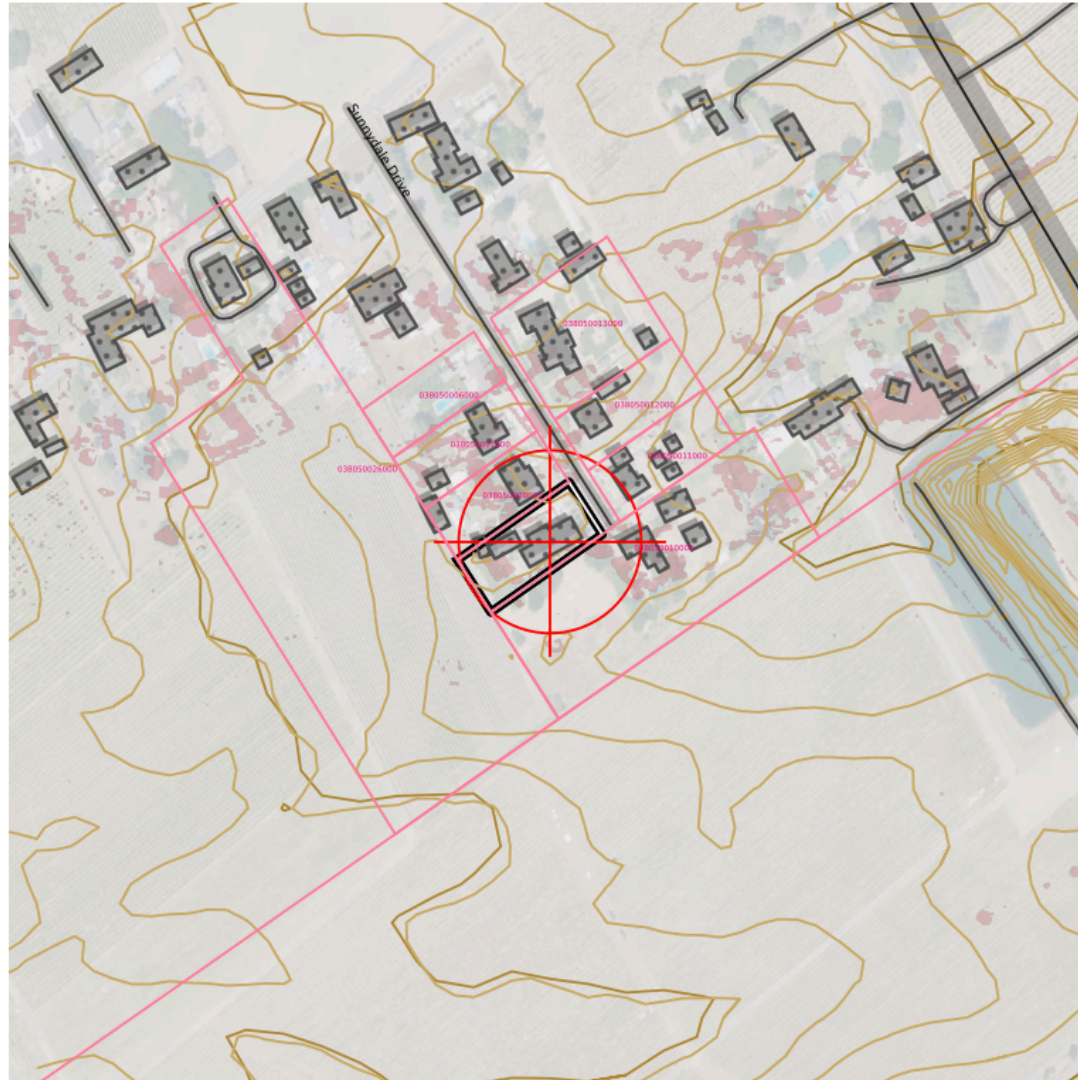


TOPOSCOUT
FIELD MAP

TopoScout Field Map
Lat: 38.34586 Lng: -122.30466
38.345860°N 122.304660°W | Radius: 30 mi

Generated: July 13, 2026
Base: USGS National Map
Generated by TopoScout - outkastdesigns.com

TOPOSCOUT SITE PLAN PREVIEW | \$19 UPGRADE FOR AUTOCAD DXF FILE



INCLUDED LAYERS



0 — 400 ft State Plane · NAD83 · US Feet

- Contours (Index/Major/Minor/1ft)
- PLSS Section/Township/QQ
- Roads Major/Local/Trail + ROW
- Buildings
- Hydro
- Power Lines
- Pipeline
- Railroad
- Benchmarks + Labels
- County Boundary
- OTLS Grid
- Site Info
- Elevation Labels
- 3D Polylines (all geometry)

✂ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – [https://www.ngs.noaa.gov/cgi-](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

[bin/datasheet.prl](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – <https://geodesy.noaa.gov/GEOID/>

GEOID18 and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

